Hazard Analysis
Software Engineering

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Table 1: Revision History

Date	**Developer(s)**	**Change**
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
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1 Introduction

MERLS is a telehealth platform for assessing the speech of bilingual children at home. Patients complete speech assessments online, and speech clinicians use the platform to review the recordings, verify the results, and manage their patients' records and analytics. MERLS streamlines the clinician's workflow by combining speech annotation and assessment analysis in one system. Annotation is supported by automatically generated transcripts that clinicians can review and correct, and analysis is supported by organized, visualized results that point out behaviors needing the clinician's attention.

Because MERLS handles sensitive health information and supports clinical decisions about children, failures in the system could have serious consequences. A hazard is defined as any condition or state of the system that, under certain circumstances, could lead to harm to patients, such as an incorrect clinical conclusion or the disclosure of sensitive information. This document identifies the hazards in MERLS, analyzes how they could arise, and describes how the system's design and requirements reduce them.

2 Scope and Purpose of Hazard Analysis

This document examines the hazards in each component of MERLS outlined below. For each hazard, the analysis determines the possible failures, along with their effects, causes, and recommended actions. It covers both technical hazards and user interaction hazards, because clinicians rely on the system's outputs when assessing a child's language ability, and the information it stores is confidential. If these hazards are not addressed, patients and clinicians could suffer losses such as the exposure of private health information, incorrect clinical conclusions that affect a child's care, the permanent loss of assessment data, and a loss of confidence in the platform.

3 System Boundaries and Components

[Dividing the system into components will help you brainstorm the hazards. You shouldn't do a full design of the components, just get a feel for the major ones. For projects that involve hardware, the components will typically include each individual piece of hardware. If your software will have a database, or an important library, these are also potential components. —SS]

The system can be separated into six major components, some of which work together to form the complete system, as described below.

3.1 ML Transcription Model

TODO

3.2 Database

The database stores all of the data that the system relies on, including patient and clinician profiles, assessment recordings, transcripts, assessment annotations and notes, and generated reports. This component also stores video recordings and sensitive health records, making it one of the most sensitive parts of the system.

3.3 Account System

TODO

3.4 Web Platform

TODO

3.5 Report Generation

This component compiles the results of an assessment into a report that clinicians can review and use when evaluating any patient. It compiles the automatically generated results and flags, along with transcription edits, annotations, and notes made by the clinician, so that the final report accurately reflects the clinician's evaluation. Once generated, reports are saved to the database and shown on the clinician's dashboard.

3.6 Video Recording

TODO

4 Critical Assumptions

[These assumptions that are made about the software or system. You should minimize the number of assumptions that remove potential hazards. For instance, you could assume a part will never fail, but it is generally better to include this potential failure mode. —SS]

- Assume that the clinicians using the system are licensed speech-language pathologists.
  - This assumption is made since the system is designed to support clinical assessment. If the clinician is not licensed, then they should not be assessing or diagnosing a child's speech and language abilities.
- Assume that the clinicians have access to an approved device with stable internet.

  - This assumption is made since the system is designed to be used on a desktop or laptop with a web browser. If the clinician doesn't have access to such a device or stable internet connection, they would not be able to use the system.

- Assume that clinicians review the system's flags and transcripts before relying on them, so that they're making the final judgement.
  - This assumption is made since the system is designed to assist clinicians, not replace them. If clinicians rely on automatically generated results without reviewing them, then errors in transcription or flagging may lead to incorrect conclusions about a child.

- Assume that patients and caregivers have consented to the assessment being recorded and stored.
  - This assumption is made since the system records and stores video of children. If consent has not been given, then the recording should not be collected or stored by the system, nor should any assessment be conducted via the system.

- Assume that the assessment contents are provided by a professional research team and are clinically correct.
  - This assumption is made since the system is designed to support real, professional clinical assessments and the team isn't responsible for designing the assessments themselves. If the assessment content is incorrect, then the results shown to clinicians may be misleading even if the system works as intended.

- Assume that assessments are conducted in languages supported by the transcription model.
  - This assumption is made since the system relies on an existing speech recognition model. If an assessment is conducted in an unsupported language, then the system will not be able to generate a meaningful transcript.

- Assume that the clinician has instructed a patient with speech or language difficulties to use this system, and that the patient is able to use the system without any assistance.
  - This assumption is made since the system is designed to be used by patients with speech or language difficulties. If the patient is not able to use the system without lots of assistance from an external body, then the assessment may not be completed accurately.

5 Failure Mode and Effect Analysis

[Include your FMEA table here. This is the most important part of this document. —SS] [The safety requirements in the table do not have to have the prefix SR. The most important thing is to show traceability to your SRS. You might trace to requirements you have already written, or you might need to add new requirements. —SS] [If no safety requirement can be devised, other mitigation strategies can be entered in the table, including strategies involving providing additional documentation, and/or test cases. —SS]

6 Safety and Security Requirements

[Newly discovered requirements. These should also be added to the SRS. (A rationale design process how and why to fake it.) —SS]

7 Roadmap

[Which safety requirements will be implemented as part of the capstone timeline? Which requirements will be implemented in the future? —SS]

8 Appendix — Generative AI Use Disclosure

8.1 AI Tools Used

- Claude Sonnet 5 (Medium)
- ChatGPT-5.6 Luna

8.2 How and Where Used

Both models were used in all the sections to do the following:

- Check that each written section answered every question asked and met the rubric requirements
- Improve the wording, clarity and the flow of written sections
- Debug LaTeX compilation errors

The models were used to do the following in specific sections

- Brainstorm possible failure modes for the system components in section 5

8.3 Verification of AI-Generated Content

To verify the output of the LLM, each team member went over their section and presented their work to the group, explaining what they had completed since the previous meeting. Each member was able to successfully explain their sections. Additionally, before any pull request was approved, a team member who had not written the section reviewed the work to ensure it was cohesive, accurate, and consistent with the rest of the document.

The factual content of the documents came from the research of the group members. The writer of each section researched, obtained, and documented all referenced sources. After the models suggested wording improvements, the writer re-read the section to confirm that it still conveyed the same point and that the research was not altered.

9 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?
4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

Question 1

Nour

What went well with this deliverable was determining possible failure modes regarding the ML Transcription model. There are several things to be wary of when dealing with external APIs and using machine learning models that have been tuned for our case. This is because they are not always predictable. Models typically have some form of evaluation, and we were given the model already chosen by our supervisors and seeing that it has not been evaluated was a bit concerning since it is hard to be certain how well such a model will perform.

What reassures me is that speech clinicians can go back and verify the model's output against the video playback, which is extremely important. However, this places the responsibility on the clinician, including how much time they are willing to spend annotating and whether they will watch the entire assessment instead of skimming through it.

Another worrying issue is data retention. If Zero Data Retention was not applied for when the team started using OpenAI's Whisper model, client information is being stored by OpenAI. OpenAI doesn't use this data to train its models unless a user explicitly agrees to share it, but the data is still stored on its servers. OpenAI is a large and popular company that could experience cyber attacks, so having sensitive data stored in multiple places is not ideal.

Catherine

As a team, hazards and risks had already been heavily discussed throughout the project, so during meetings, we were able to easily brainstorm and discuss each section of the deliverable, and divvy up all the parts. As we now had quite a bit more resources, it was also easier to ask more guided questions when we were confused or not aligned with one another on a concept related to our project, so communication was also a lot more clear for this deliverable. Pre-splitting the system into components also made it easy to divide the work between us, since each person could focus on the parts they were most familiar with.

Question 2

This deliverable was relatively straightforward, and our team did not run into any major issues while completing it. We began by brainstorming the components of the system together, and we were able to identify what could go wrong with each one and possible hazards that could result. MERLS is a privacy-critical system, so security and data breaches were extremely important for us to address. The system also depends on performance and quality for different features to work, which made it easy to determine its high-level components. Because we share a strong understanding of the system, prioritizing the safety requirements was not difficult. Our supervisors also mentioned which parts of the system they wanted completed first, which gave us a clear direction as to what were the most important parts of the system.

Question 3

Before starting this deliverable, we had already thought about and discussed in pretty deep detail different risks and hazards to consider, especially for a project that touches real patients and health practices. For instance, as mentioned throughout the deliverable, because the system stores video recordings and health records of children, privacy and data security were something we talked about extensively from the start. They also come up in our previous deliverables. We also knew that the transcription model could have trouble differentiating between speakers, different languages, or sentences in general, so we also knew that a clinician would always have to be within the loop and ensure that everything has been reviewed and verified by a licensed clinician.

Other risks only came up while doing the Hazard Analysis, where going through each component made us think about how each part could fail on

its own, instead of just the system as a whole. For example, while writing about the database and report generation module, we realized a clinician could potentially view or edit the wrong patient assessment, or lose important edits without knowing, resulting in incorrect results. Writing the critical assumptions also brought up risks we hadn't considered, like what happens if the internet cuts out partway through an assessment, or if the child happens to need a lot of help from a parent during their assessment and the parent happens to interfere too much. Trying to decide whether these should be assumptions or actual hazards made us think more about what we could realistically control, and if so then how.